Research Summary Sheet - Article 3

Complete the following form for each artifact that validates your problem statement.

Source (APA format and include a link to the source)

Waltz, W. (2023). *LANDO: Developing autonomous operations for planetary surfaces NASA Advisory Council Technology, innovation, and engineering committee*.

https://www.nasa.gov/wp-content/uploads/2024/05/waltz-eci-lando-tagged.pdf?emrc=6a085a04e51e9

Artifact Summary

This team developed LANDO, which is an autonomous system for handling/offloading lunar payloads. The system is designed to detect payload characteristics, safely lift objects, and place them onto mobile vehicles to transport. It addresses the problem of autonomous cargo operations. This connects to my HUNCH project since the majority of the robot is autonomous cargo handling.

Artifact Critique

This article is relevant since it focuses on autonomous lunar cargo handling. For my HUNCH project, the strongest part would be the information on the practical problems faced with the robot, such as payload detection, safely lifting the cargo, and titled lander decks. However, LANDO deals more with the cargo handling itself as opposed to transporting it long distance.

Reliability

The presenter, Walter Waltz, along with other researchers Matthew Vaughn and Iok Wong, each have multiple research publications to their name on aerospace development and spacecraft transportation. The project lead, Julia Cline, is a NASA Langley aerospace engineer with a PhD. The work was developed through NASA's Space Technology Mission Directorate and presented at the AIAA SciTech Forum, making it a very reliable source.